A02

FOUNDATION

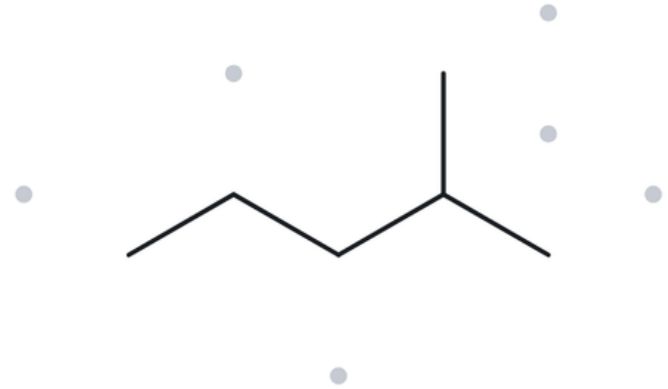
Parent?
> Two potential parent chains exist, but they are identical; all of 5 carbons.
> Parent chain is pentane.

Numbering?
> From left, substituents are on 4C.
From right, 2C.
> Therefore, count from the right.

Substituent/s?
> Substituent is 2-methyl.

Identity?
> 2-methylpentane.

AlkaneDraw?
> Correct!



2-methylpentane

A02

FOUNDATION

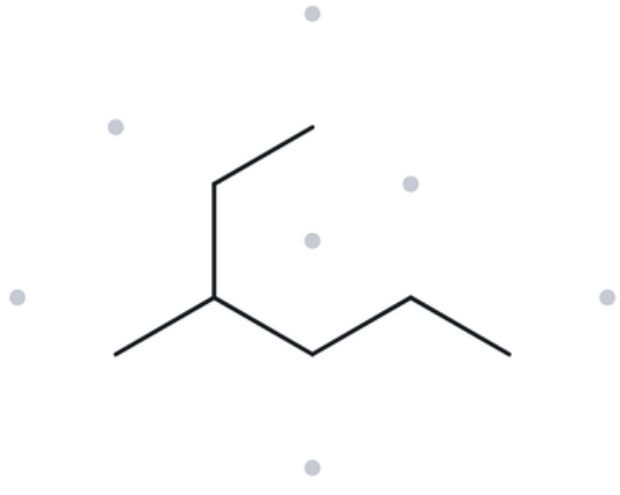
Parent?
> Longest chain is 6 carbons long, and not at all in a straight line.
> Therefore, parent chain is hexane.

Numbering?
> From left, substituents are on 3C.
From right, 4C.
> Therefore, count from the left.

Substituent/s?
> Substituent is 3-methyl.

Identity?
> 3-methylhexane.

AlkaneDraw?
> Correct!



3-methylhexane

A03

FOUNDATION

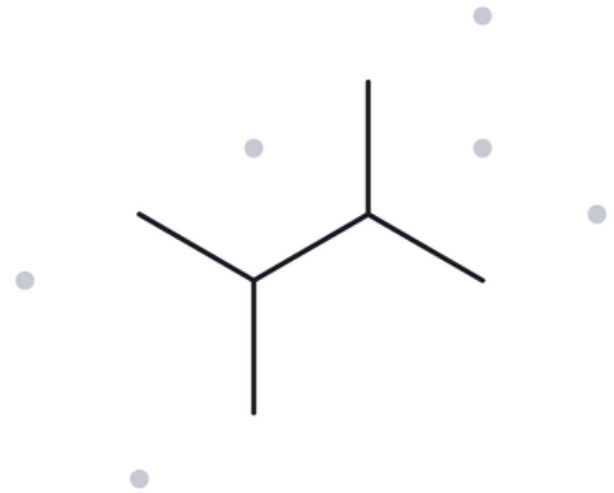
Parent?
> Longest chain is 4 carbons long. On inspection we find that there are 4 identical ways to trace this.
> Therefore, parent chain is butane.

Numbering?
> From both left and right, substituents are on 2C and 3C.
> Therefore, count from either side.

Substituent/s?
> Two methyl groups exist at 2,3.
> Thus, substituents are 2,3-dimethyl.

Identity?
> 2,3-dimethylbutane.

AlkaneDraw?
> Correct!



2,3-dimethylbutane

A04

DEVELOPING

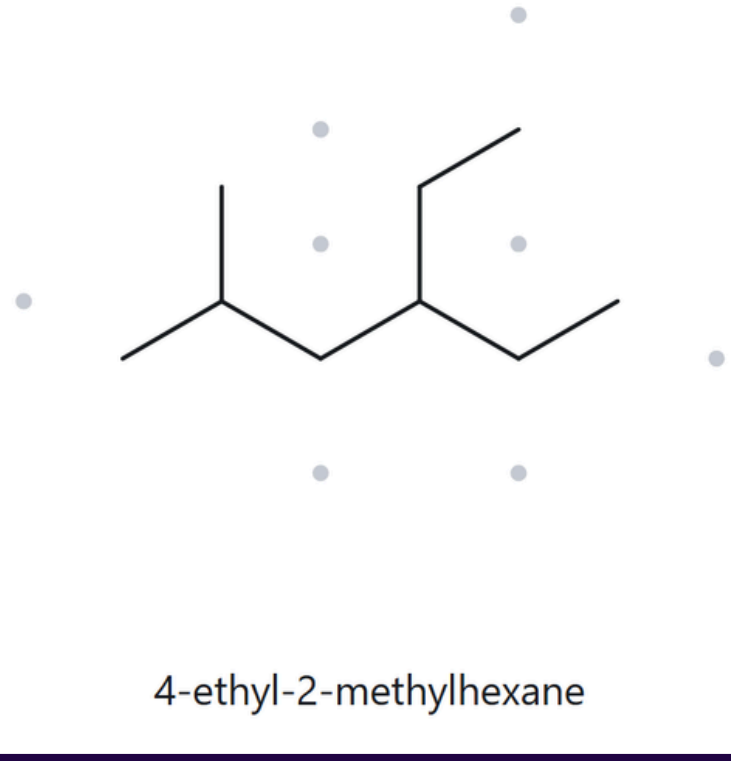
Parent?
> Longest chain is 6 carbons long. On inspection we find that there are 4 identical ways to trace this.
> Therefore, parent chain is hexane.

Numbering?
> From left, substituents are on 2C and 4C. From right, 3C and 5C. Count from left.

Substituent/s?
> Methyl at 2, ethyl at 4. Using IUPAC's alphabetical rule, substituents are 4-ethyl-2-methyl.

Identity?
> 4-ethyl-2-methylhexane.

AlkaneDraw?
> Correct! (Note the sketch limitation.)



4-ethyl-2-methylhexane

A05

DEVELOPING

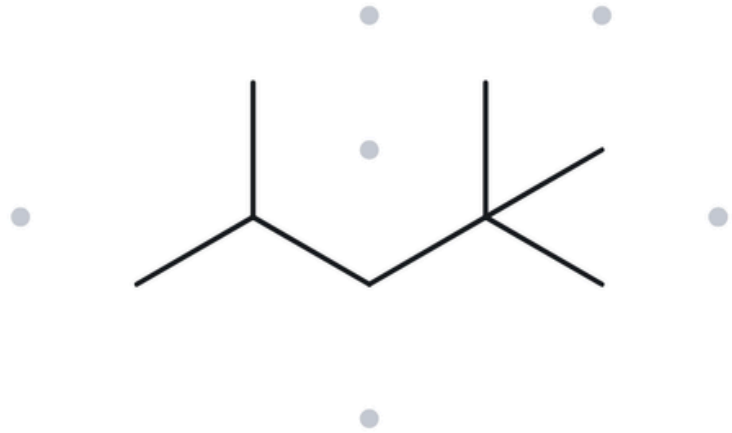
Parent?
> Longest chain is 5 carbons long. On inspection we find that there are 6 identical ways to trace this.
> Therefore, parent chain is pentane.

Numbering?
> From left, 1 substituent is on 2C and two are on 4C. From right, the pair are on 2C. Count from right.

Substituent/s?
> Methyl pair at 2, lone methyls at 4. Three methyls --- 2,2,4-trimethyl.

Identity?
> 2,2,4-trimethylpentane.

AlkaneDraw?
> Correct! (Note the sketch limitation.)



2,2,4-trimethylpentane

A06

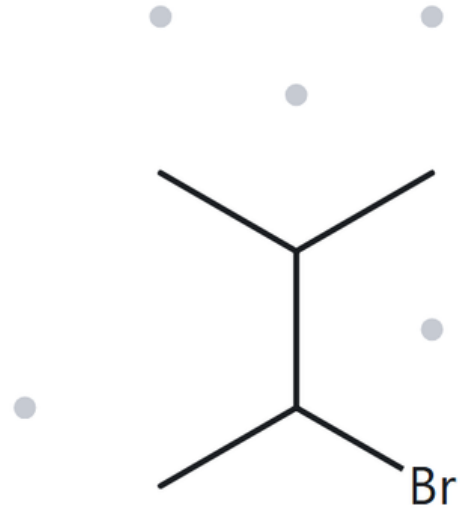
DEVELOPING

Parent?
> Longest chain is of 4 carbons, traced along 2 identical paths
> Therefore, parent chain is butane.

Numbering and Substituents?
> Present is a primary alkyl halide group: R-C(-X)-R' , with bromine; and a methyl group. The halide takes priority.
> From left, the priority group is at 3C, but from right it is at 2C. Thus, we count from the right.
> Substituents are 2-bromo-3-methyl.

Identity?
> 2-bromo-3-methylbutane.

AlkaneDraw?
> Correct!



2-bromo-3-methylbutane

A07

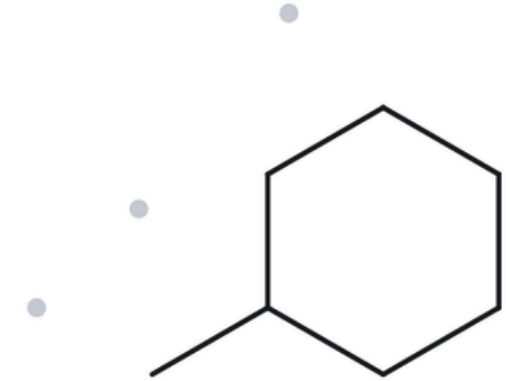
DEVELOPING

Parent?
> Longest chain is of 6 carbons in a ring.
> Therefore, parent chain is cyclohexane.

Numbering and Substituents?
> Only substituent is a methyl group.
> Because parent chain is a ring, we begin counting wherever the priority substituent is, labelling its attachment atom as 1C.
> Substituent is 1-methyl or simply, methyl.

Identity?
> methylcyclohexane.

AlkaneDraw?
> Correct!



methylcyclohexane

A08

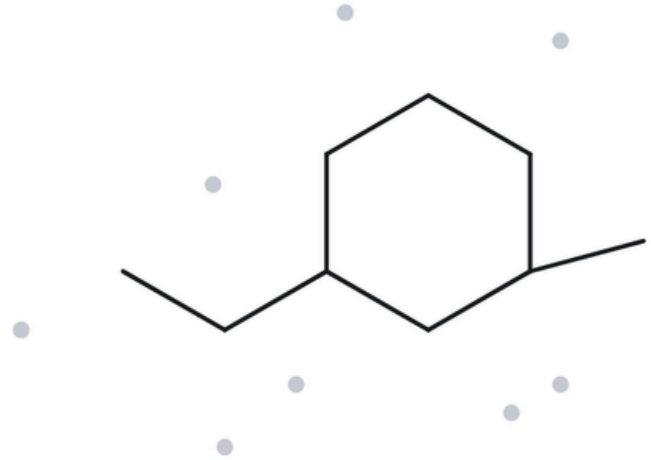
CHALLENGE

Parent?
> Longest chain is 6 carbons long, either through the linear chain or the cyclic ring. Cyclic ring takes precedence.
> Parent chain is cyclohexane.

Numbering and Substituents?
> Same as in A07, we start wherever the priority group is.
> There are an ethyl group and a methyl group. Priority is first alphabetically.
> Going clockwise from ethyl gives us the other group at 5C. Counterclockwise, substituents are 1-ethyl-3-methyl.

Identity?
> 1-ethyl-3-methylcyclohexane.

AlkaneDraw?
> Correct! (Note the sketch limitation.)



1-ethyl-3-methylcyclohexane

A09

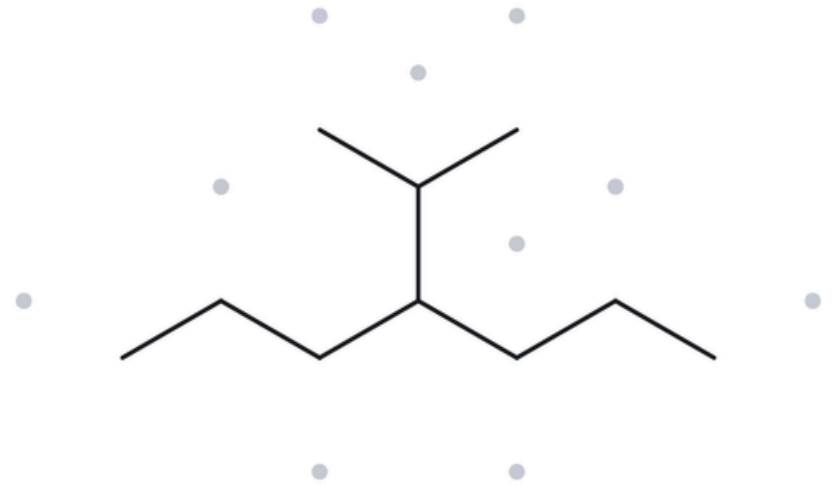
CHALLENGE

Parent?
> Longest chain is 7 carbons long, but the chain with the greatest number of attachments is 6 carbons long.
> Parent chain is hexane.

Numbering and Substituents?
> There are a methyl group and a propyl group. Methyl takes priority in counting.
> Taking this into account, methyl is closest to an endpoint at 2C, which places propyl at 3C.
> Substituents are 2-methyl-3-propylhexane.

Identity?
> 2-methyl-3-propylhexane.

AlkaneDraw?
> Correct! (Note the sketch limitation.)



2-methyl-3-propylhexane

A10

CHALLENGE

Parent?
> Longest chain is linear with 4 carbons, but the 3-carbon ring, as always with rings, takes precedence.
> Thus, parent chain is cyclopropane.

Numbering and Substituents?
> With cyclopropane as the parent chain, only one substituent exists: the lone butyl we had rejected as the parent chain.
> Substituents are 1-butyl or simply, butyl.

Identity?
> butylcyclopropane.

TheAlkane?
> Correct! (Note the sketch limitation.)



butylcyclopropane